TEXORA · COURSE NOTES · LECTURE SHEET

Applications of Differential Geometry

The geometry of curves measured with calculus — arc length, curvature, the radius and centre of curvature, evolutes, and torsion for curves in space.

MATH-1101 · MATHEMATICS-I · TOPIC 3 · LEVEL 1 TERM 1

1. Arc length

Differential geometry asks calculus questions about curves: how long is this piece of curve, how sharply does it bend here, and where is the centre of the circle that best fits it? The natural measure along a curve is the **arc length** s, measured from some fixed point of the curve. For a plane curve y = f(x),

$$s = \int \sqrt{1 + (dy/dx)^2}\, dx, \quad \text{so that} \quad ds/dx = \sqrt{1 + (dy/dx)^2}.$$

For a curve given parametrically as x = x(t), y = y(t), the integrand becomes $\sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$; for polar curves r = r(θ), it becomes $\sqrt{r^2 + (dr/d\theta)^2}\, d\theta$. Measuring position by arc length rather than by x or t gives the *intrinsic* description of a curve, independent of how it happens to be drawn on axes.

WORKED EXAMPLE — ARC LENGTH

Find the length of $y = (2/3)\, x^{3/2}$ from x = 0 to x = 3. Here dy/dx = √x, so ds/dx = √(1 + x) and $s = \int_0^3 \sqrt{1 + x}\, dx = (2/3)\, [(1 + x)^{3/2}]_0^3 = (2/3)\,(8 - 1) =$ **14/3**.

2. Curvature

Curvature measures how fast a curve turns. If ψ is the angle the tangent makes with the x-axis, the curvature at a point is the rate of turning per unit arc,

$$\kappa = |d\psi/ds|, \quad \text{and for } y = f(x): \quad \kappa = |y''| / (1 + y'^2)^{3/2}.$$

A straight line has κ = 0 everywhere (it never turns). A circle of radius a has constant curvature κ = 1/a — small circles bend sharply, large ones gently, which is why curvature is compared against circles. The parametric form is $\kappa = |x'y'' - y'x''| / (x'^2 + y'^2)^{3/2}$.

2.1 Radius and centre of curvature

The **radius of curvature** $\rho = 1/\kappa$ is the radius of the **circle of curvature** (osculating circle) — the circle that touches the curve at the point and matches its bending there. Its centre, the **centre of curvature**, lies on the normal to the curve at distance ρ from the point, on the concave side. For $y = f(x)$ the centre $(\bar{x}, \bar{y})$ is

$$\bar{x} = x - y'(1 + y'^2)/y'', \quad \bar{y} = y + (1 + y'^2)/y''.$$

WORKED EXAMPLE — CURVATURE OF A PARABOLA

For $y = x^2$: $y' = 2x$ and $y'' = 2$, so $\kappa = 2 / (1 + 4x^2)^{3/2}$. At the vertex ($x = 0$) the curvature is greatest: $\kappa = 2$, hence $\rho = 1/2$, and the centre of curvature is $(0, 1/2)$ — the circle of radius 1/2 sitting on the vertex and matching the parabola's bend there. Far from the vertex the parabola straightens and κ falls toward 0.

3. Evolutes and involutes

As one travels along a curve, the centre of curvature moves too, and its path is a new curve called the **evolute** of the original. Running the relationship backwards, the original curve is an **involute** of its evolute: an involute is traced by the end of a taut string being unwound from the evolute. Involutes are not a curiosity — the profiles of most gear teeth are involutes of a circle, because that shape makes mating gears roll together with constant velocity ratio. The evolute of a parabola is a semicubical parabola; the evolute of a circle is a single point, its centre.

4. Curves in space: the Frenet frame and torsion

A space curve needs two measures instead of one. At each point, three mutually perpendicular unit vectors form the **Frenet frame**: the **tangent** T (direction of motion), the **principal normal** N (direction in which the tangent is turning), and the **binormal** $B = T \times N$. **Curvature** $\kappa = |dT/ds|$ still measures bending within the curve's instantaneous plane, while **torsion** τ measures how the curve twists out of that plane — a plane curve has $\tau = 0$ identically, and a helix has both κ and τ constant. The rates of change of the frame are summarised by the **Serret–Frenet formulas**:

$$dT/ds = \kappa N, \quad dN/ds = \tau B - \kappa T, \quad dB/ds = -\tau N.$$

Together, κ and τ as functions of arc length determine a space curve completely (up to its position in space) — the fundamental theorem of curve theory, and the three-dimensional completion of this topic.

5. Why an engineer cares

Curvature is the mathematics of bending, and bending is everywhere in engineering: the profile of a cam or gear tooth, the path of a road or railway through a curve (where curvature sets safe speed), the bending of yarns and fibres around guides and needles in textile machinery, and the drape of a fabric, which is governed by how its yarns curve. The formulas of this topic are the tools for putting numbers on "how sharply does it bend here?"

6. Intrinsic equations — describing a curve from inside

Since κ is a function of arc length for any curve, the relation $\kappa = \kappa(s)$ — or, in older form, s as a function of the tangent angle ψ — is called the curve's **intrinsic equation**: it describes the curve completely without any coordinate axes. A straight line is s arbitrary with $\kappa = 0$; a circle of radius a is $\kappa = 1/a$ (constant); the logarithmic spiral has κ proportional to 1/s. Two curves with the same intrinsic equation are the same curve, merely placed differently in the plane — which is why this topic's quantities, not the coordinates, are the geometry.

7. Key points and summary

- Arc length accumulates as $ds = \sqrt{(1 + y'^2)}\, dx$ (with parametric and polar variants), and position along a curve is measured intrinsically by s.
- Curvature $\kappa = |y''| / (1 + y'^2)^{3/2}$ measures turning per unit arc; $\rho = 1/\kappa$ is the radius of the osculating circle, whose centre is the centre of curvature.
- The locus of centres of curvature is the evolute; unwinding a string from a curve traces its involute — the shape of gear teeth.
- Space curves need the Frenet frame and a second invariant, torsion τ, measuring twist out of the osculating plane; κ and τ together determine the curve.

8. Practice

Work each problem fully; final answers are given at the end.

1. Find the curvature of $y = x^2$ at the point $x = 1$, and the radius of curvature there.
2. A circle has radius 3. State its curvature and radius of curvature at every point.
3. Find the length of $y = (2/3)\, x^{3/2}$ from $x = 0$ to $x = 1$.
4. State the centre of curvature of $y = x^2$ at its vertex (use §2.1's result).
5. Explain, in one or two sentences, why every plane curve has torsion zero.
6. A road bend is modelled by a circle of radius 80 m. What curvature must the road's geometry be built to?

Answers

1. $\kappa = 2 / (1 + 4)^{3/2} = 2 / (5\sqrt{5}) \approx 0.179$; $\rho = 1/\kappa \approx 5.59$.
2. $\kappa = 1/3$ everywhere; $\rho = 3$ (its own radius, as it must be).
3. $s = (2/3)\,[(1 + x)^{3/2}]_0^1 = (2/3)\,(2\sqrt{2} - 1) \approx 1.22$.
4. $(0,\ 1/2)$.
5. Torsion measures twisting out of the osculating plane; a plane curve never leaves its plane, so there is nothing to measure — $\tau \equiv 0$.
6. $\kappa = 1/80 = 0.0125\ \text{m}^{-1}$.
